



Evaluation of Groundwater Hydrochemistry in Egbeagu Amansea, Anambra State, Nigeria for Sustainable Water Management

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Abstract: Rapid urbanization in Egbeagu Amansea of Nigeria poses a significant threat to the maintenance of groundwater quality, thus creating a requisite to support effective water management with comprehensive data. This study investigated the hydro-chemical characteristics of groundwater in Awka North, Anambra State. Samples of groundwater were collected from seven boreholes and a hand-dug well during the wet season. These samples were analyzed for physiochemical parameters, such as pH, electrical conductivity (EC), total dissolved solids, total hardness, major cations (Ca^{2+} , Mg^{2+} , Na^+ , and K^+), and anions (HCO_3^- , Cl^- , SO_4^{2-} , and NO_3^-). The study employed standard hydro-chemical methods, such as Piper and the United States salinity (USSS) diagrams to characterize water types and determine the dominant hydro-chemical processes influencing groundwater chemistry. The results of the Piper trilinear diagram revealed that bicarbonate ($\text{HCO}_3^- + \text{CO}_3^{2-}$) was the dominant anion, hence reflecting carbonate dissolution in the aquifer. Sodium adsorption ratio (SAR) values ranged from 0.53–0.674, thus classifying all samples in the low (S1) category and indicating minimal sodium hazard for soil. EC values spanned 44–130.6 $\mu\text{S}/\text{cm}$, placing samples in the low (C1) to medium (C2) categories. The study confirms that the groundwater in the study area is suitable for drinking and irrigation purposes.

Keywords: Groundwater quality; Hydrochemical facies; Sodium adsorption ratio; Water quality index; Piper trilinear diagram

1 Introduction

Groundwater, one of the largest freshwater resources, serves as the primary provision of drinking water globally [1]. Given the high population density in urban areas, there is likelihood of groundwater contamination [2], which could subsequently threaten the health of thousands of people. Consequently, management of water resources and its related studies have placed a high priority on providing potable and palatable water [3]. Therefore, it is crucial to determine the status of groundwater quality and evaluate its suitability for drinking [4]. Assessing groundwater quality ensures compliance with standards such as the World Health Organization (WHO) Guidelines for Drinking-Water Quality [5] and the Nigerian Industrial Standard [6].

Groundwater chemistry is shaped by natural processes and human activities during its movement from recharge to discharge zones [7–9]. These processes include mineral dissolution, ion exchange, and anthropogenic contamination from activities such as improper waste disposal, excessive extraction of groundwater, and agricultural runoff with high concentration of fertiliser [10, 11]. Moreover, leachates from pit latrines and septic systems can infiltrate and contaminate groundwater sources, hence compromising their quality [12].

The hydrochemistry of groundwater is also influenced by chemical properties of recharged water, reactions between water and minerals, rock-water interactions, geochemical formations, and intermixing of water [13]. For instance, while soil horizons act as natural filters attenuating contaminant migration, elevated concentrations of metals could occur if minerals in the earth crust leach into groundwater [14]. The geology and geographic configuration of an area affect the inherent chemical quality of groundwater [15]. Hydrochemical evaluations of groundwater flow systems are generally based on the availability of data on groundwater chemistry [16].

Groundwater serves as a critical source of potable water within the Egbeagu Amansea. However, excessive extraction of groundwater in the region poses potential threats to the quality and sustainability of these groundwater

resources. From the literature, there is no study on the hydrochemical characteristics of the groundwater in this specific area. This lack of information hinders effective groundwater management and exposes the population to potential health risks associated with contaminated water. This study aims at evaluating the hydrochemical characteristics of groundwater in the above-mentioned region.

2 Materials and Method

2.1 Study Area

The study was conducted in a small village of Egbeagu Amansea, Awka North local government of Anambra State, Nigeria. Egbeagu Amansea is among the five villages in Amansea rural community and lies approximately between latitude $06^{\circ} 16' 16.91631''$ N and longitude $07^{\circ} 07' 34.01516''$ E. The inhabitants of this community rely mainly on their stream, known as Amaowelle, for domestic activities. The stream ($06^{\circ} 15' 05.86739''$ N, and $07^{\circ} 08' 21.68037''$ E) is located at the lower course of Ezu river [17]; in this connection, few commercial boreholes were drilled in the strategic locations within the area. The region experiences a tropical savannah climate with distinctly wet and dry seasons, receiving an average annual rainfall of approximately 1,800 mm. Geologically, Awka North is underlain by the Imo shale formation and the Bende-Ameki formation, which influence the hydrogeological characteristics of the groundwater system. The map of the study area is shown in Figure 1.

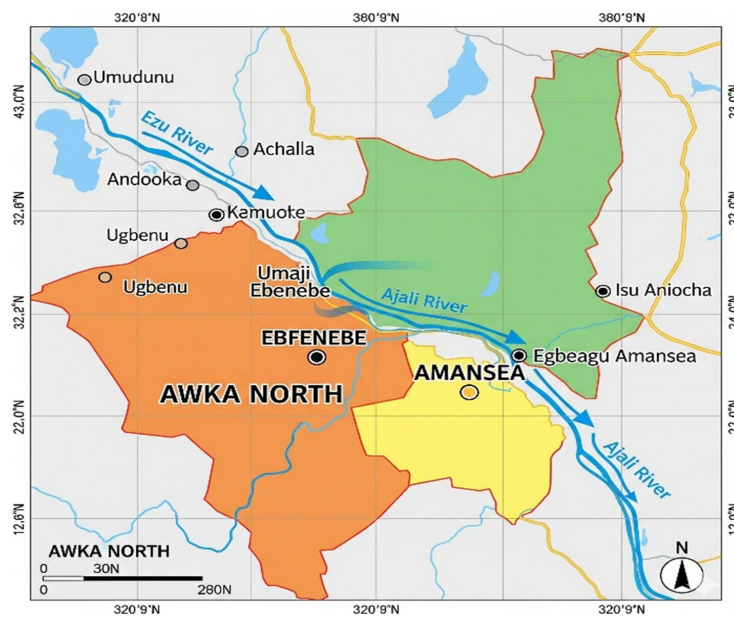


Figure 1. Geographical location of Egbeagu Amansea community

2.2 Sample Collection

A total of eight groundwater samples were collected from seven boreholes (BH1, BH2, BH3, BH4, BH5, BH6, and BH7) and one hand-dug well (WL3) located in Egbeagu Amansea. The boreholes and the well were purged by pumping water for 5–10 minutes to ensure representative groundwater was collected. Sterile 1-liter polyethylene bottles were rinsed with the groundwater to be sampled before collection. Two sets of samples were collected at each point: One for physicochemical analysis (pH, electrical conductivity (EC), total dissolved solids (TDS), total hardness, carbonate, fluoride, nitrate, sulphate, and chloride) and another for cation analysis (calcium, potassium, sodium, and magnesium). Samples for cation analysis were acidified with 2–3 drops of concentrated nitric acid (HNO_3) to a $\text{pH} < 2$ to preserve metal ions. Samples were labelled with identification number, date, and time, and stored in a portable cooler at 4°C for analysis within 24–48 hours to minimize degradation.

2.3 Groundwater Sampling Strategy

To ensure balanced coverage of the entire small village, seven boreholes and a hand-dug well (at close proximity to BH3) were purposively selected to represent aquifer types in the area.

This study was intended to be a preliminary assessment of the groundwater in the Ebeagu-Amansea community. As shown in the map of Figure 1, Egbeagu is geographically a confluence settlement. Samples were collected in the wet season between June and July 2025 to account for either infiltration of polluted surface runoff to the hand-dug well or intrusion from nearby rivers into the boreholes. Sampling was done by the grab method, which involved

collecting a single sample from each source at a given time, an approach considered reliable for assessing physical, chemical, and microbiological properties of the drinking water [18]. At the well and boreholes, bottles were rinsed three times before sampling; water was allowed to flow for about 2–3 minutes before collection.

2.4 Laboratory Analysis

The groundwater samples were analysed for the following parameters: pH, EC, TDS, total hardness, calcium (Ca^{2+}), potassium (K^+), sodium (Na^+), magnesium (Mg^{2+}), carbonate (CO_3^{2-}), fluoride (F^-), nitrate (NO_3^-), sulphate (SO_4^{2-}), and chloride (Cl^-). Analytical methods followed standard procedures outlined by the American Public Health Association (APHA) [18]. The measurements of the cations and anions were performed using a 752N UV-VIS spectrophotometer and spectra AA 220||S atomic absorption spectrophotometer (AAS), respectively.

2.5 Data Analysis

The hydrochemical data were analysed using descriptive and multivariate statistical techniques, such as the Grapher software and Microsoft Excel (version 2016). Hydrochemical facies were assessed using Piper's diagram and United State salinity (USSL) diagram based on milliequivalent ion concentrations (Ca^{2+} , Mg^{2+} , Na^+ , K^+ , HCO_3^- derived from CO_3^{2-} , Cl^- , and SO_4^{2-}).

Descriptive statistics (mean, standard deviation, minimum, and maximum) were calculated for each parameter to summarize the qualities of groundwater. Results were compared with the WHO guidelines and Nigerian Standard for Drinking Water Quality (NSDWQ) to assess their suitability for drinking and other purposes. Duplicate samples were analysed to assess reproducibility, with relative standard deviation (RSD) maintained below 5%. The ion balance error was calculated for each sample, with an acceptable range of $\pm 5\%$.

$$\text{The SAR was calculated using the formula: } \text{SAR} = \frac{\text{Na}^+}{\sqrt{\frac{\text{Ca}^{2+} + \text{Mg}^{2+}}{2}}} \quad (1)$$

where,

SAR = Sodium adsorption ratio;

Na^+ = Sodium ion;

Ca^{2+} = Calcium ion;

Mg^{2+} = Magnesium ion.

The water quality index (WQI) was calculated using the formula:

$$\text{WQI} = \frac{\sum_{i=1}^n w_i q_i}{\sum_{i=1}^n w_i} \quad (2)$$

where, w_i = unit weight of i^{th} water quality parameter:

$$w_i = \frac{k}{s_i} \quad (3)$$

$$\text{where } k = \frac{1}{\sum_{i=1}^n S_i} \quad (4)$$

q_i = quality rating (sub-index) of i^{th} water quality parameter and is given as:

$$q_i = 100 \left(\frac{v_i - v_{io}}{s_i - s_{io}} \right) \quad (5)$$

where,

v_i = estimated value of i th water quality parameter;

v_{io} = idea value of i th water quality parameter;

s_i = standard permissible value of the i th water quality parameter;

In most cases, $v_{io} = 0$, except for pH and dissolved oxygen (DO);

For pH, $v_{io} = 7$, for DO = 14.6 mg/l.

The rating of water quality using the above Eqs. (2)–(5) is shown in Table 1.

Table 1. Ratings of water quality for various water quality index (WQI)

WQI	Rating of Water Quality
0–25	Excellent
26–50	Good
51–75	Poor
76–100	Very poor
Above 100	Unsuitable for drinking

Source: Adopted from Elemile et al. [19]

3 Results and Discussion

3.1 Physiochemical Parameters

The mean pH of 7.52 indicated slight alkaline that fell within the standards stipulated by WHO (6.5–8.5) and NSDWQ (6.5–8.5). EC ranged from 36.7 to 185.8 $\mu\text{S}/\text{cm}$ (mean 96.31 $\mu\text{S}/\text{cm}$), suggesting low mineralization below the WHO and NSDWQ limits of 1000 $\mu\text{S}/\text{cm}$ (Table 2). TDS varied from 56.2 to 773 mg/L (mean 66.53 mg/L), also below the 500 mg/L guideline, and indicated good water quality. Total hardness averaged 101.23 mg/L CaCO_3 , with a range of 74.2 to 126 mg/L to classify the water as soft to moderately hard. Major cations showed mean concentrations of 6.24 mg/L for Ca^{2+} , 4.97 mg/L for Mg^{2+} , 8.23 mg/L for Na^+ , and 8.5 mg/L for K^+ , all well within WHO and NSDWQ limits (75, 50, 200, and 12 mg/L, respectively). Anions included a mean SO_4^{2-} of 31.06 mg/L (range of 10–49 mg/L), Cl^- of 115.63 mg/L (range of 90–150 mg/L), NO_3^- of 5.48 mg/L (range of 3.14–7.62 mg/L), F^- of 0.19 mg/L (range of 0.16–0.2 mg/L), and CO_3^{2-} of 166.88 mg/L (107.5–195 mg/L). In this regard, all anion levels were below the respective WHO and NSDWQ limits. Figure 2 presents a plot of the physiochemical parameters of the groundwater samples.

Table 2. Summary of the water quality in the boreholes and well under study

Parameter	Min	Max	Mean	SD	World Health Organization (WHO) Limit [20]	Nigerian Standard for Drinking Water Quality (NSDWQ) Limit [6]
pH	7.12	7.92	7.523	0.267	6.5–8.5	6.5–8.5
Electrical conductivity (EC) ($\mu\text{S}/\text{cm}$)	36.7	185.8	96.312	53.332	1,000	1,000
Total dissolved solids (TDS) (mg/L)	56.2	77.3	66.538	7.400	500	500
Total hardness (mg/L as CaCO_3)	74.2	126	101.225	17.477	500	150
Calcium (Ca^{2+}) (mg/L)	5.167	8.289	6.238	1.249	75	75
Magnesium (Mg^{2+}) (mg/L)	5.016	5.417	4.973	0.387	50	20
Sodium (Na^+) (mg/L)	7.487	9.167	8.232	0.568	200	200
Potassium (K^+) (mg/L)	7.478	9.567	8.497	0.706	12	12
Sulphate (SO_4^{2-}) (mg/L)	10	49	31.056	12.477	250	100
Chloride (Cl^-) (mg/L)	90	150	115.625	17.896	250	250
Nitrate (NO_3^-) (mg/L)	3.143	7.762	5.477	1.939	50	50
Fluoride (F^-) (mg/L)	0.163	0.206	0.190	0.014	1.5	1.5
Carbonate (CO_3^{2-}) (mg/L)	107.5	195	166.875	38.353	-	-

Hydrochemical facies can be classified on the basis of dominant ions using the Piper trilinear diagram. The concentrations of major ionic constituents of groundwater samples were plotted in the Piper trilinear diagram to determine the water types (Figure 3) and illustrate the hydrochemical compositions of the eight groundwater samples. This graphical representation, comprising two basal triangles (for cations and anions) and a central diamond field, facilitates the classification of water types and inference of geochemical processes. From the cationic triangular fields of the Piper diagram, it was observed that 100% of groundwater samples fell into the no dominant type. The anion triangle revealed bicarbonate ($\text{HCO}_3^- + \text{CO}_3^{2-}$) as the dominant anion (70–90%), reflecting carbonate dissolution in the aquifer. According to Ali and Ali [21], the concentration of bicarbonate and carbonate also affected the suitability of water for irrigation purpose. It has been hypothesized that all Ca^{2+} and Mg^{2+} precipitate as carbonate. This was further proposed by Eaton [22] on the concept of residual sodium carbonate (RSC) for the evaluation of high carbonate waters. Though bicarbonate aids digestion in human body and is found to have no known adverse effects

on human health, it might cause kidney stones when it exceeds 300 mg/l in the drinking water and in the presence of higher concentration of calcium during dry climatic conditions [21]. The dominant bicarbonate and carbonate ions suggest natural influences, possibly due to the interaction of natural water-rock.

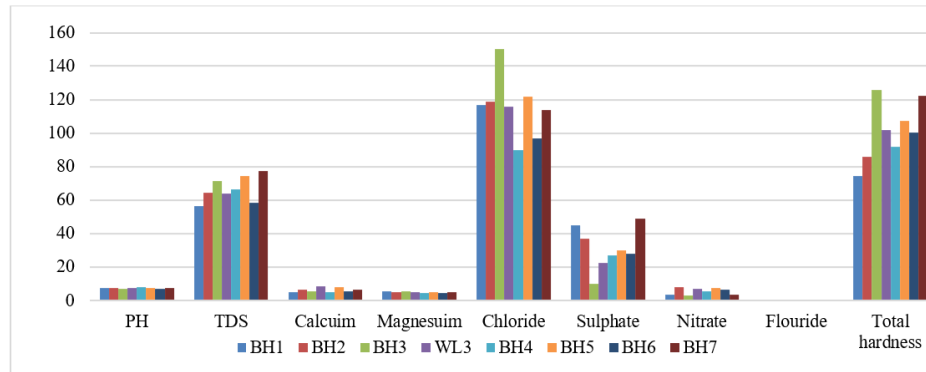


Figure 2. Plot of water quality parameters from the sampling boreholes and well

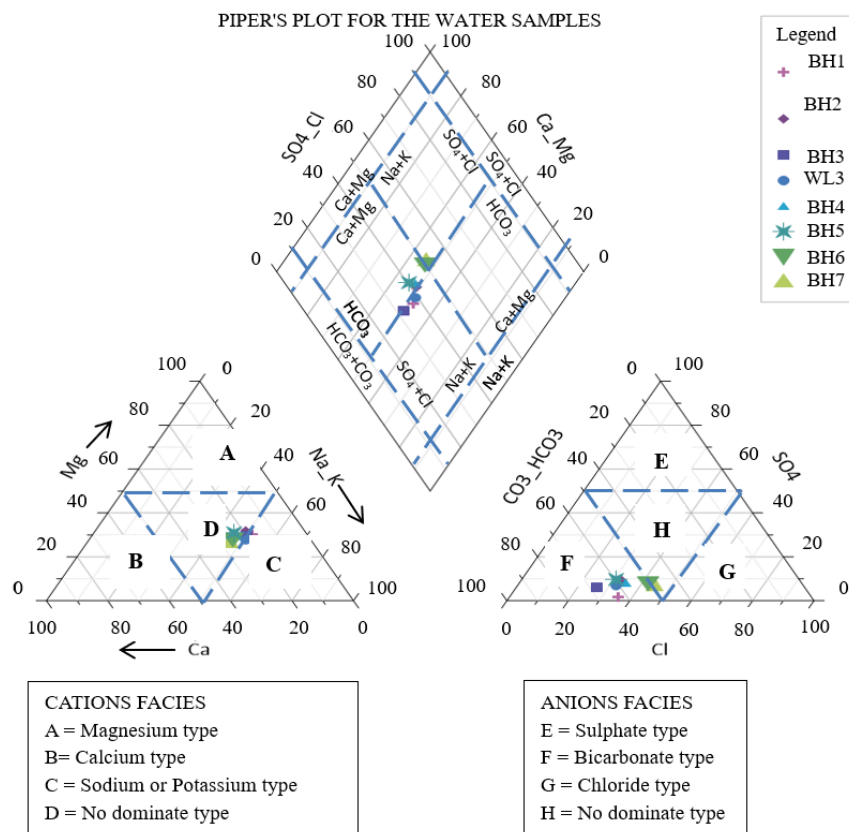


Figure 3. Piper trilinear diagram of water from the sampling well and boreholes

The other 10% of the groundwater samples fell into no dominant type. The plot of chemical data on diamond shaped trilinear diagram revealed that both alkaline earth and alkaline metal were the most dominant metals in the water samples of the study area. From the data plots, it is apparent that the total hydrochemistry is dominated by alkaline earths. The absence of samples in Na-K-Cl or saline fields underscores the character of the freshwater and rules out significant seawater intrusion or dissolution. These findings imply a geologically homogeneous aquifer system, with carbonate weathering as the primary control of water chemistry. The dominance of good quality water in this season suggests that rainfall recharge helps dilute contaminants, thereby improving groundwater quality. It also highlights relatively favorable aquifer conditions within the region.

USSL diagram in Figure 4 classifies the irrigation suitability of the eight groundwater samples (BH1, BH2, BH3, BH4, BH5, BH6, BH7, and WL3) based on SAR and EC (Table 3). The SAR values ranged from approximately 1

to 5, classifying all samples in the low (S1) category and indicating minimal sodium hazard for soil permeability. EC values spanned 250–1000 $\mu\text{S}/\text{cm}$, thus placing samples in the low (C1) to medium (C2) categories, with BH1, BH2, BH3, BH4, BH5, and WL3 near 250 $\mu\text{S}/\text{cm}$ (C1); BH6 and BH7 near 750–1000 $\mu\text{S}/\text{cm}$ (C2). This suggests low to moderate salinity risk, which is suitable for most crops with good drainage. All samples fell within the S1-C1 to S1-C2 zones, indicating excellent to good irrigation water quality (Table 4).

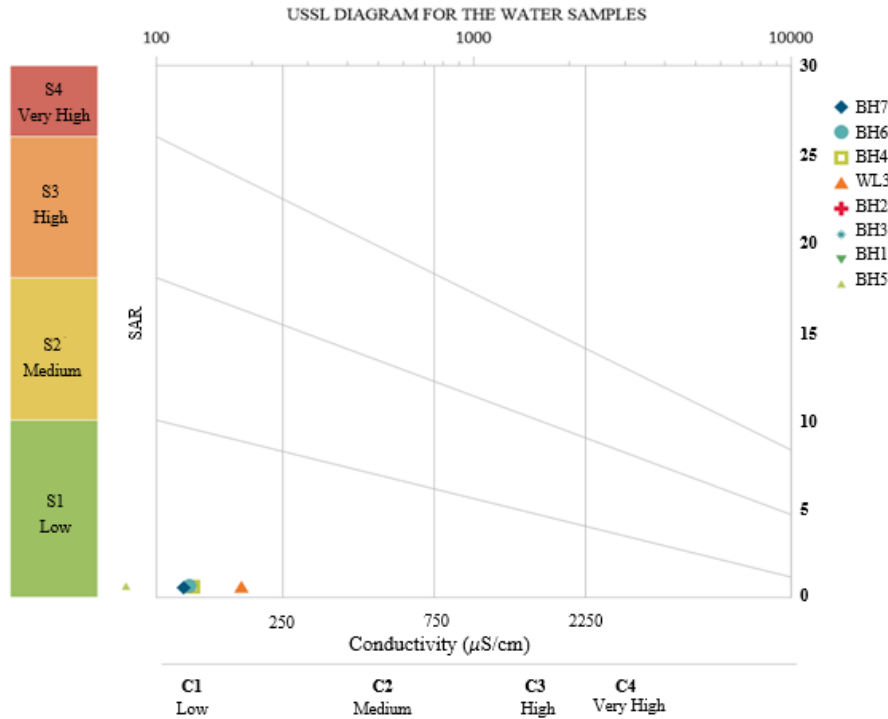


Figure 4. United States salinity (USSL) diagram of the water samples from the sampling well and boreholes

Table 3. Sodium adsorption ratio (SAR) and electrical conductivity (EC) values of water from the sampling well and boreholes

Sample ID	Sampling Well and Boreholes	EC (micromhos/cm)	SAR
BH1	Borehole 1	45.2	0.612
BH2	Borehole 2	36.7	0.547
BH3	Borehole 3	44	0.670
WL3	Hand-dug well 3	185.8	0.588
BH4	Borehole 4	130.6	0.606
BH5	Borehole 5	80	0.564
BH6	Borehole 6	126.7	0.674
BH7	Borehole 7	121.5	0.530
	Mean	96.3	0.599

The WQI was calculated using pH, TDS, total hardness, Ca^{2+} , Mg^{2+} , Cl^- , SO_4^{2-} , and NO_3^- , with weights assigned based on their significance to health. The mean WQI derived from the summarized data is approximately 17, thus indicating “Excellent” quality (<50) according to the classification (<50: Excellent, 50–100: Good, 100–200: Poor, >200: Unsuitable). All parameters were within the WHO and NSDWQ limits. The “Excellent” WQI (Figure 4) confirms that groundwater in Egbeagu Amansea is suitable for drinking and irrigation.

4 Conclusions

In the study, the consistently low TDS, EC, and WQI values indicated that the groundwater was of excellent quality for drinking. All sampled locations also met the WHO and NSDWQ standards and served as well-preserved aquifers with limited pollution. The dominance of the HCO_3^- facies and the low levels of most ions were due to natural processes (e.g., carbonate dissolution). The C1-S1 classification in the USSL diagram confirmed that the

Table 4. Ratings of the water quality of samples collected from the boreholes and well

Sampling Well and Boreholes	Sample Code	Water Quality Index (WQI)	Rating of Water Quality Recommended by United Nations Environment Programme (UNEP) [23]
Bolehole 1	BH1	14.925	Excellent
Bolehole 2	BH2	16.594	Excellent
Bolehole 3	BH3	17.087	Excellent
Hand-dug well 3	WL3	16.446	Excellent
Bolehole 4	BH4	16.296	Excellent
Bolehole 5	BH5	16.569	Excellent
Bolehole 6	BH6	16.565	Excellent
Bolehole 7	BH7	15.802	Excellent

groundwater was suitable for irrigation, and the low SAR values further indicated that the soil structure was unlikely to be adversely affected by this water when used for irrigation.

In conclusion, good water quality dominated the wet season due to recharge effects. Groundwater in this study area was considered safe without pollution from sewage discharge, industrial effluent discharge, and dense settlement. Hence, additional development and extraction could still be supported by the private sector, such as establishing the bottled-water industry for boosting the economic and sustainable utilization of groundwater resources in this locality. Moreover, the growth of food production in the state could be achieved by the government via extending the irrigated fields in this community.

Author Contributions

Conceptualization, I.I.N.; methodology, I.I.N.; software, I.I.N.; validation, I.I.N. and J.C.O.; formal analysis, I.I.N.; investigation, I.I.N.; resources, I.I.N. and J.C.O.; data curation, I.I.N.; writing original draft preparation, I.I.N.; writing review and editing, I.I.N. and J.C.O.; visualization, J.C.O.; supervision, I.I.N.; project administration, I.I.N. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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